

FORECASTING DNF DECISIONS

If you have been keeping track during this chapter, I've used the dreadful DNF rates at the Leadville Trail 100 as an example of poor goal setting. There's also much to be learned from observing how and where athletes make the choice to drop out. Every year I make it a point to spend several hours at the Outward Bound aid station at the Leadville Trail 100. Located at mile 75 of the race, it is a DNF decision point for many. Runners come into the aid station in all different states. Some look great, others awful. Despite their varying outward-looking conditions, there is no real correlation between the athletes' appearances and who drops out or continues on to the finish. There is, however, an internal calculation made by each and every person as they enter that aid station. Whether they realize it or not, they look at how they feel at the moment and extrapolate that to how they anticipate feeling while running the remaining twenty-five miles back to the town of Leadville—which include the infamous Powerline climb. According to the perceived exertion end point interaction concept, if the slope or the end point of that imaginary line exceeds what they perceive is tolerable to them, they get their wristband cut, head into their car, and take the short ride back to town. The athletes that forge on have imaginary lines that are within their preconceived limitations, whether through a higher exertion tolerance, a more accurate forecast, or a lower degree of instantaneous suffering. However, the potential for error in both of these strategies remains the same: you are forecasting for a long period of time with many unknowns.

IN THE MOMENT

The take-home message for ultrarunners is that there's some method to the madness of "being in the moment." Forecasting and predicting in ultrarunning can be a fool's errand, although we're naturally inclined to do it. We're even trained to incorporate perceived exertion end point interactions in workouts and hard runs. However, unlike in a workout or marathon, your ability to predict how you will feel hours down an ultramarathon trail is clouded by the sheer duration and your impaired ability to comprehend (because let's get real: are you smarter or dumber after eighty miles?). So, when you get to your own Outward Bound aid station and all seems hopeless, forget the forecasting. Stay in the moment, focus on your effort at the present time, and draw upon some of the mental skills that we covered in Chapter 15. The finish line will eventually show up—just give it time.

Sleep Considerations for Longer Events (200 miles, etc.)

Ultramarathons lasting more than 48 hours require some special planning, particularly in regard to sleep. This is an area where the research is pretty thin, and we are doing a lot of educated guesswork in terms of when, where, and how much to sleep during events. Rather than just leave it at "it depends," let's go over the basics of what we know about when, where, and why the sleep monster might jump on your back.

THE WITCHING HOUR

Ultrarunning folklore is riddled with tales of hallucinations, out-of-body experiences, paranormal activities, and even an alien sighting or two. Given enough time and sleep deprivation, ghouls and goblins, devils and demons, witches and wildlife will inevitably appear, haunting even the heartiest of souls. I've been there. Deep into the Tor des Géants I swore some random Italian at an aid station was one of my lifelong friends. The bewildered look on

his face after I gave him a big giant bear hug was the only thing that snapped me back to the fact that I was on a mountain ledge at 8,000 ft. I've also seen it firsthand, like when an athlete I was pacing at the Badwater 135 was having a full-blown conversation with the Michelin Man, complete with an origin story and advice on electrolytes. But do you ever wonder why all of these tales of the supernatural happen around the same time? Why do the Care Bears magically appear at 3 a.m. and not 3 p.m.? Why is it called the “witching hour” and not the “witching ambiguous period in the middle of the night?” The fact is, the vast majority of these silly stories and tall tales occur during a very narrow window of time, between 2 and 4 a.m., and as it turns out, that’s no cosmic coincidence.

The term “witching hour” actually has roots hundreds of years old. Many years ago, it was thought that the separation between the living and the dead was narrowest somewhere between the hours of 2 and 4 a.m. This thought became so prevalent that in the 1500s the Catholic Church actually banned activities during this time for fear of witches and warlocks ushering the dead into the living domain.

In ultrarunning terms, the phenomenon is similar. At some point in the wee hours of the morning, weird stuff starts to happen. Rocks turn into bears. Branches move and take the form of talking snakes. And more importantly, your performance declines.

A 2020 study published in the journal *Sleep Science* looked at runners during the Fort Clinch 100 Mile Endurance Run and tracked their pace and dropout rates during the race, as well as their sleeping habits during the race (Brager et al. 2020). I was not surprised to learn the 100-mile runners had their slowest paces and highest dropout rates at exact same time, at about 3:30 a.m. (which is denoted by Zeitgeber time 20 on the x-axis in Figure 16.11).

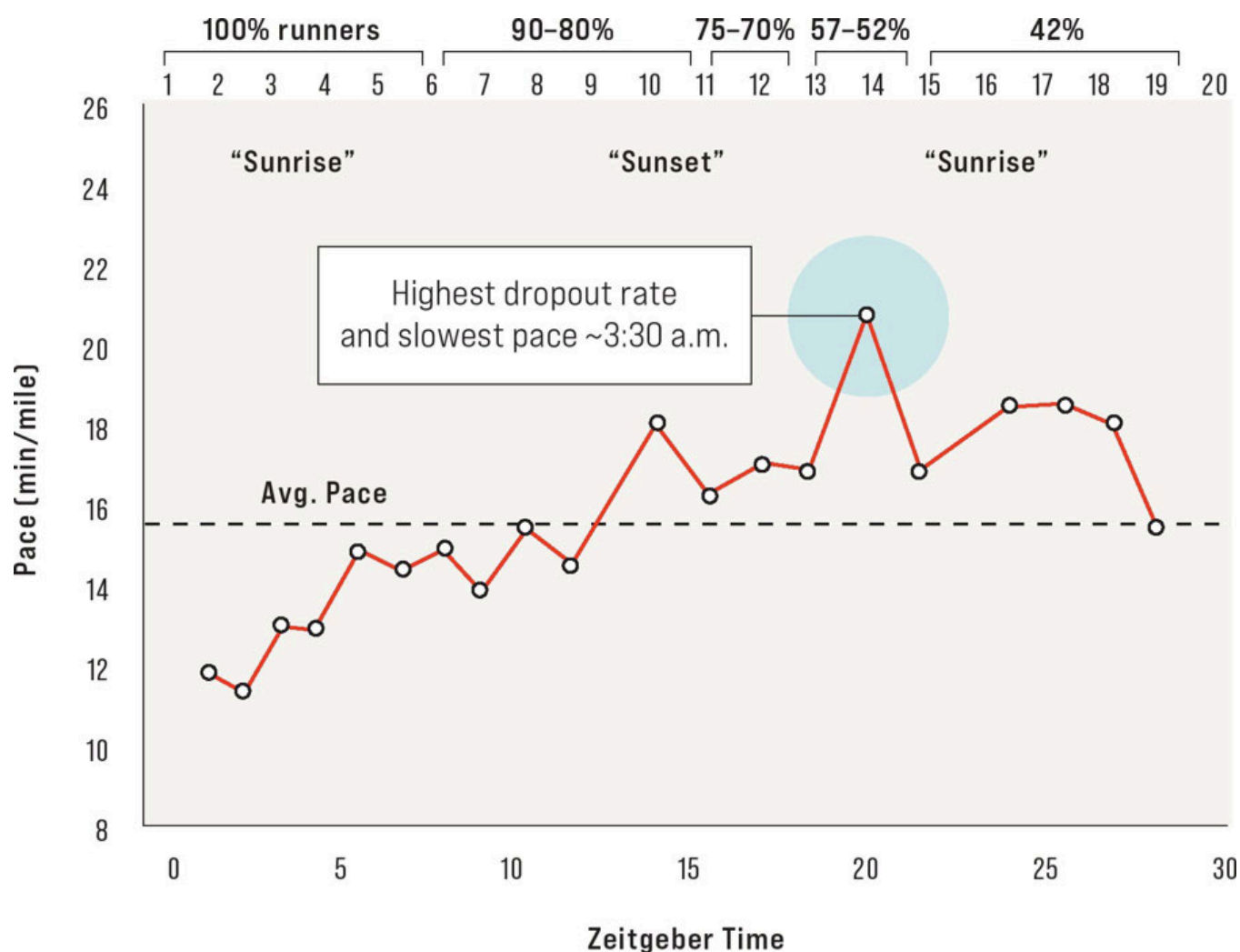


FIGURE 16.11 When athletes drop out in a 100-mile race. Source: Brager et al. 2020.

This aligns with research looking at circadian rhythms showing that some people have more difficulty controlling core temperature when required to exercise in the early hours of the morning (Muginshtein-Simkovitch et al. 2015). So, there's some science to the fact that hallucinations and the sleep monster seem to show up for everyone at the same time.

SLEEP EXTENSION

Athletes try all manner of training strategies to avoid going zero miles an hour while curled up in a sleeping bag during a race. Some try overnight runs, while others simply stay up for a couple of days in a row, powering through life with a few extra cups of coffee. Many feel that the experience of being a new parent better prepares them for sleep deprivation. This begs the question: can you actually improve your tolerance for sleep deprivation? Meaning, is there some physiological improvement you can make in training that causes your body to adapt and perform better (or less badly) when sleep-deprived?

The answer is a resounding no. You can't train sleep deprivation simply by going on an overnight run, the way you train a muscle to be stronger by lifting heavier weights. The aforementioned strategies are experiential—they will help you get accustomed to and familiar with the sensation of sleep deprivation, which can be helpful, but it's hard to tell if those experiences actually improve future performances.

Therefore, I advocate that athletes only do one or two overnight runs at the very most, and sometimes not at all. The toll you pay for these bouts of sleep deprivation is high, and the benefit is slim to none.

While you might not be able to “train” sleep deprivation in a classic way, you can undertake strategies in advance of an overnight event to help mitigate the negative consequences. Sleep extension (sometimes referred to as sleep banking) is a simple and effective way to improve any performance where sleep deprivation might become an issue. Researchers have studied this strategy for decades and have found that, in general, one week of extending sleep by as little as an hour per night can improve cognitive function and performance. Additionally, and more relevant for longer ultramarathons, sleep extension appears to have a positive effect on performance during bouts of sleep deprivation that occur immediately after the extension, particularly from a cognitive perspective. This is great news for anyone going into a twenty-four-plus-hour event where decision-making, navigation, and mood will be important for success.

HOW TO DO IT

Fortunately, sleep extension is easy. Most research protocols have centered around a one- to two-week sleep extension protocol where total time in bed is ten hours per night (Vitale et al. 2019). If the time commitment is problematic for you, don't fret—there does appear to be a linear dose response for sleep extension protocols up to ten hours. This means that if you habitually sleep seven hours, even an extension of one hour per night for one week will have some positive effect.

While I would love for every athlete I work with to be able to sleep ten hours a night, let's face it, life happens. From a practical standpoint, I've had my athletes target nine to ten hours a night for two weeks leading up to a big overnight race. When adding sleep hours, it's preferable (and usually the most practical), but not necessary, to go to sleep earlier as opposed to waking up later. Nine hours of sleep a night for one week seems to have a moderate and tangible effect on their ability to stay awake and alert during an overnight event, provided they weren't chronically sleep-deprived to start out with. This strategy also leverages the efficacy of the sleep extension without being too intrusive on their daily schedule, so it's a win all the way around.

SLEEPING DURING A RACE

When and how long to sleep during races is one of those areas where few people agree. In addition, there's not a lot in the literature we can draw from to inform best practices. We do know that sleep restriction (withholding sleep for a few hours for at least a night) can negatively impact endurance performance by as much as 4 percent. Total sleep deprivation can impair performance by over 11 percent (Cullen et al. 2019; Fullagar et al. 2015). Taking a nap might help curb some of that deterioration, but it is unclear to what extent and how much it would apply during an ultramarathon. So, the general working philosophy I use with my athletes breaks races up into three segments:

1. RACES SHORTER THAN FORTY-EIGHT HOURS: NO SLEEP NEEDED

For these races, there might be no sleep needed, provided that you performed the sleep extension protocol mentioned earlier and you respond to short-term sleep deprivation normally.

2. RACES FROM FORTY-EIGHT TO SEVENTY-TWO HOURS: SHORT NAPS

For these races, I advise athletes to get short 20- to 40-minute trail-side naps (or in an aid station if it's convenient) close to sunrise on the second day, and each morning after. The theory (and this is just a theory pioneered by adventure racers) is that timing when you wake up to right when the sun rises will align with your body's natural circadian rhythms.

3. RACES LONGER THAN SEVENTY-TWO HOURS: 90- TO 120-MINUTE CHUNKS OF SLEEP

For races lasting longer than three days, I advise runners to get at least 90 to 120 minutes of sleep, beginning during the second night. For races of this length, very few people can remain awake for the entirety of the event. And even for those who can, performance normally deteriorates to the point where some sleep provides more than enough of a benefit to make up for the time not moving.

In general, for most athletes I advise sleeping more than less, as long as you have the time to make cutoffs. While this might increase your downtime at aid stations, you are better able to preserve your performance on the trail, sometimes dramatically (10 percent or more if you read into the literature). Being extremely sleep-deprived can cost you hours on the racecourse due to deteriorated performance, bad decision-making, forgotten nutrition, and even a fall. Getting some sleep might cost you time in the short run but will pay off later down the trail.

Creating a Race-Day Nutrition Strategy

If you have done everything right in training, you should not need an elaborate raceday nutrition plan because your race nutrition will be very similar to what you ate in training. A simple plan is just as effective and is much easier to remember and execute as compared to a complex plan. I've seen inordinately detailed plans and have always likened them to the biochemistry experiments I performed as a young student. In these unnecessarily complicated plans, the required food and fluid are detailed in thirty-, twenty-, and sometimes even ten-minute increments. Puzzled and confused crew members waded through the details, asking, "Was this aid station supposed to be 8 oz of Coke with a Nutella cracker, or water with half a scoop of the red powder, half a scoop of the other powder, and a peanut butter and jelly sandwich?" When watching these real-life chemistry experiments unfold, I have always wondered, "Is this how that runner trains?" To date, in all my experience as a runner and a coach, I

have yet to come across an athlete who maintains such elaborate nutrition planning during training. Why, then, do they do it in a race?

THE NAPPUCCINO

IF YOU ANTICIPATE SLEEPING at any point during the event, researchers have discovered a great strategy involving caffeine and short power naps. While the findings are not specific to an ultramarathon setting, they can be modified and applied if you anticipate needing sleep or if an unanticipated trail nap is a must. In these situations, if you expect to take a twenty- to forty-minute power nap, it is best to consume caffeine just prior to the slumber. Paradoxically, the crux of the benefit lies in the time it takes for the stimulant to take effect. When you wake up, you'll be more refreshed and alert (Hayashi, Masuda, and Hori 2003; Horne and Reyner 1996; Reyner and Horne 1997).

[Table 16.1 presents an example of such an overcomplicated nutrition plan. I adapted it from many of the flawed race plans I have seen over the years. Although the plan is well intended, it is too complicated, and the concoctions utilized are impractical to replicate in training. I once read a nutrition plan developed for an athlete that included consuming exactly 83 oz of liquid during a twenty-one-mile section of race. Not 82, not 84, but 83.](#)

DEVELOP A STRATEGY, THEN A PLAN

You don't want to leave your nutrition to chance in an ultramarathon, but what you want to develop is a race-day nutrition strategy, after which you may or may not choose to create a race-day nutrition plan. A strategy encompasses the target calories, fluid, and foodstuffs you have tried and tested in training. If you develop an actual written-down-on-paper plan from that overarching strategy, it should take no more than an elementary school education to execute. If you have done a good job in training by experimenting with the right foodstuffs and drinks, the race-day strategy should naturally end up being simple.

RACE SECTION	FOOD	FLUID	SUPPLEMENTS
Start to aid station 1	2 gels	1 drink mix in bottle 1	1 salt tab
		1 drink mix in bottle 2	1 amino acid capsule
Aid station 1 to aid station 2	2 gels	Water in bottle 1	1 salt tab
	1 energy bar	1 electrolyte tablet in bottle 2	
		Coke in aid station	
Aid station 2 to aid station 3	1 pack energy chews	1/2 drink mix, 1/2 scoop whey protein in bottle 1	2 salt tabs
	1/2 pack energy chews	Water in bottle 2	
		Ginger ale in aid station	
Aid station 3 to finish	2 gels	1/2 drink mix, 1/2 Coke in bottle 1	1 salt tab
	1 pack energy chews	1 electrolyte tablet in bottle 2	1 amino acid capsule

TABLE 16.1 An overcomplicated nutrition plan.

Developing your nutrition strategy for race day should revolve around the following aspects:

- Eat and drink what you have tried in training.
- Target calories per hour.
- Target fluid per hour, adjusted for the temperature range of the event.
- Know when and how to incorporate things like caffeine or stomach-calming tricks like antacids and ginger.

FLUIDS FIRST. As discussed in Chapter 13, your nutrition strategy is largely dependent on your hydration status. When your hydration status is good, the foods you ingest stand a chance of being absorbed and metabolized into energy. Therefore, your race-day nutrition strategy starts with the fluid you plan to take in during various parts of the race. Consider factors such as time between aid stations, temperature, and intensity. If you plan to consume liquid with some form of calories, this also sets the beginning of your caloric range, which you need to determine next.

FIND YOUR TARGET CALORIE RANGE. As opposed to a strict “eat this at thirty minutes, eat that at sixty minutes” type of plan, I recommend first targeting a specific calorie range. Once that range is determined, the specific foodstuffs you plan to consume on race day are developed during day-to-day training. The calorie count is relatively standard: 200 to 300 calories per hour after the first sixty minutes for most people. Some elite athletes can push that up toward 400 calories per hour. This process is described in Chapter 13.

DEVELOP YOUR BULL’S-EYE AND OUTER-RING FOODS. During training, particularly during longer runs, use your target calorie ranges and then experiment with different foodstuffs. This is how you develop the core group of bull’s-eye foods that are tried and true for you. As described in Chapter 14, you also need to develop foods that fit into the outer rings of your target—those you can use for backup if your core go-to foods begin to fail or aren’t available. It is important to build your lists of target foods and foods that are off target, so you can make good choices in any race, under varying circumstances, and as your preferences change during an event.

Once the core foods have been fully tested in training, experiment with your backup foods. In theory, the three to five bull’s-eye foodstuffs developed in training will be all you need, but in ultrarunning, things don’t always go as planned. These outer-ring foodstuffs are what you can fall back on when all else fails. The typical fallback plan revolves around the aid station fare of cookies, soup, fruit, sandwiches, and the like.

An example of an athlete’s bull’s-eye nutrition strategy is shown in Figure 16.12. I recommend trying to stay on target and rotating through your three to five bull’s-eye foods for as long as you can during the race. If you get tired of those foods, then move to the outer rings.

CUSTOMIZE INTAKE FOR YOUR RACE DISTANCE AND EFFORT. Your blood flow is always being balanced between digestion, cooling, and your working muscles. Different intensities will allow you to digest different combinations and amounts of your bull's-eye foodstuffs. In general, the higher the intensity of the race, the more you will need to rely on easily digestible calories. The converse is also true. The longer the race, the less the intensity, and therefore the more you can incorporate complex foods. The complete "food target" (bull's-eye plus outer rings) represents any of the things you can eat, not necessarily the specific or optimal ones for a particular race. For example, you might only use gels from your bull's-eye in a 50K, gels and pretzels for a 50-mile race, and the entire rotation for a 100-mile event.

Figure 16.13 depicts a target that is customized for a more intense race. Using only three foodstuffs, this athlete would still have a real food (rice ball) and an engineered food (gel), all while including sweet, salty, and savory taste profiles.

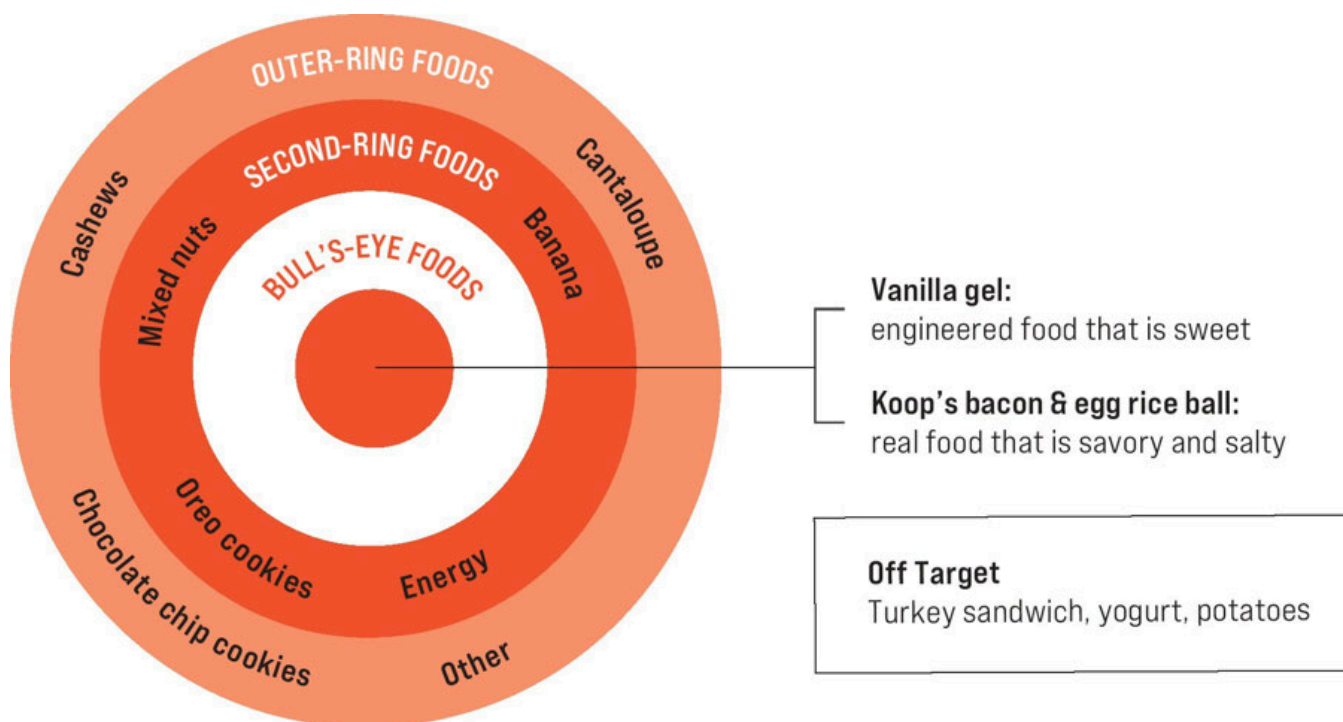


FIGURE 16.12 Example of a bull's-eye nutrition plan.

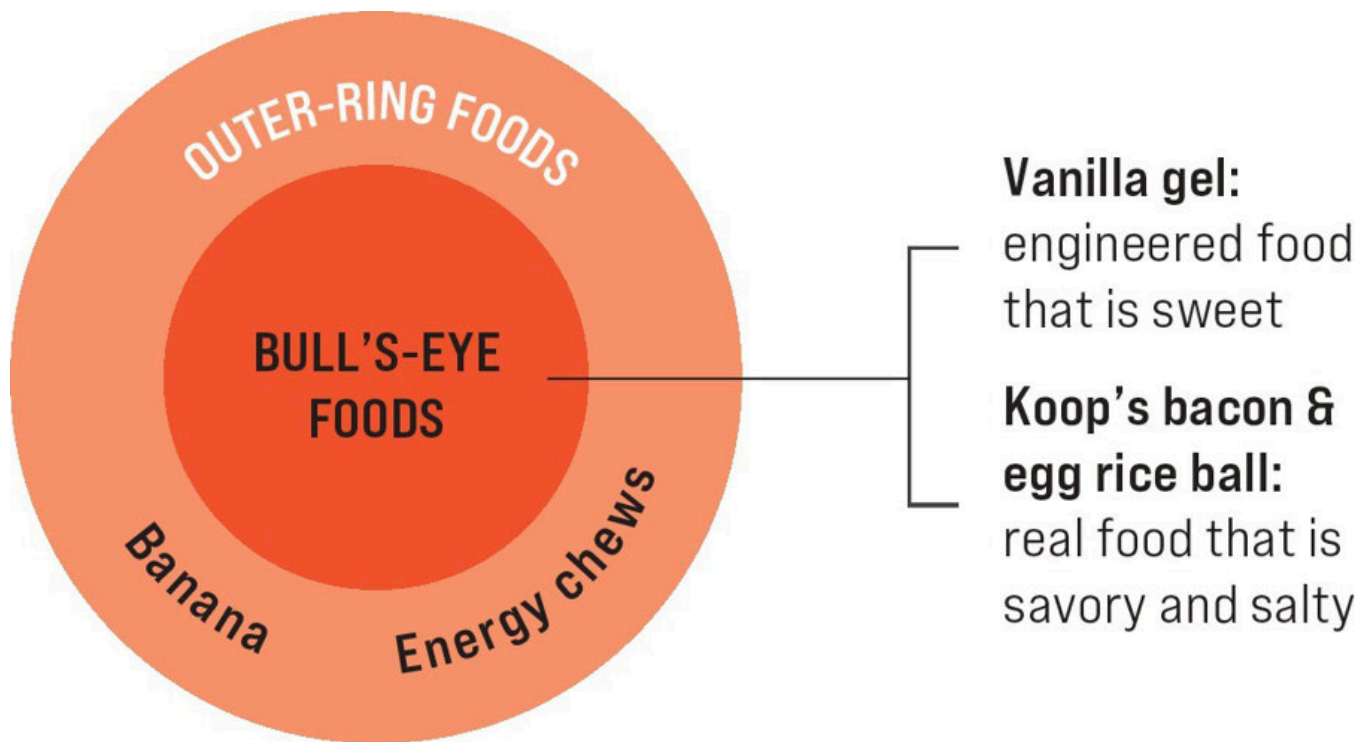


FIGURE 16.13 Target customization for a shorter, more intense ultra.

SUPPLEMENTS AND ERGOGENIC AIDS

Like bull's-eye nutrition, other supplements and ergogenic aids you anticipate using on race day should be tested in training. Yes, this means taking a caffeine pill or some crystallized ginger on a four-hour run when it is completely unnecessary! Your bull's eye nutrition might be enough to get you to the finish line on race day. But because the distance and duration of the event are typically longer than any single training run, it's good to have a set of supplements and ergogenic aids that you can go to if (or when) the need arises.

Caffeine: The Original Performance-Enhancing Drug

Yes, caffeine is a drug. In fact, it is the most widely consumed psychoactive drug in the world today. Yes, it can enhance your performance. I vividly remember pacing my wife during the latter stages of the Run Rabbit Run 100 and stumbling upon a ream of a dozen 200 mg caffeine pills. I wondered whether the runner who dropped the package was simply being lazy by not cutting the foil pack down into more realistic chunks of a few pills or actually intended to use all 2,400 mg of caffeine over the last twenty miles of the race. Because ultrarunners are a somewhat obsessive group, I assumed the latter. That would have been a sight! I'm glad I found the pills, and not an amped-up arrhythmic runner.

Caffeine supplementation can be used in two ways: to acutely enhance your performance or as a stimulant specifically to stay awake and alert.

CAFFEINE AS A PERFORMANCE ENHANCER

Caffeine supplementation starts with your morning cup of coffee on race day. If you regularly drink a cup or two a day, feel free to enjoy a similar amount as you go through your pre-race routine. Supplementation starts from this baseline, based on the race distance you are about to undertake.

For shorter races lasting less than six hours, you can supplement with occasional caffeine, up to 50 mg per hour, for the entire race. The supplementation should mainly come from caffeinated gels, chews, tablets, and colas. If you typically have coffee in the morning, caffeine from the coffee should be enough for the first two to three hours of the race. Therefore, I suggest waiting until after the second or third hour to start supplementing with caffeine in other forms. Furthermore, the dose response from caffeine as it relates to endurance performance is not linear, meaning that moderate doses of caffeine are likely to have the same performance effect as higher doses (Graham and Spriet 1995; Pasman et al. 1995). More isn't always better. Thus, a cautious and conservative approach will likely be preferable to a more aggressive one.

CAFFEINE TO STAY AWAKE AND ALERT

Many ultrarunning events go into the night and through the following day. Runners are constantly battling the mythical-yet-real sleep monster, particularly in the wee hours of the morning, before sunrise and after twenty hours on their feet. Caffeine is one of the key pieces of ammunition against the sleep monster, whether you ingest it in the form of pills, colas, chocolate-covered espresso beans, energy drinks, or caffeinated sports nutrition products. However, when using caffeine as a stimulant to boost your alertness, the timing matters more than the form.

For athletes competing in events lasting longer than twenty-four hours, caffeine is best viewed as a stimulant to stay awake and alert. In these situations, you need to focus on when the stimulation will be needed most and then supplement at that point. This means starting the day with a normal routine, including pre-race breakfast and coffee if that is what you are accustomed to. As the race begins, however, take care to avoid caffeine. Gels, drinks, and foods at this point should all be noncaffeinated. Sometime after midnight, when you expect a visit from the sleep monster, begin your caffeine supplementation. Doses can be as high as 100 mg/hr for three to four hours and can cease shortly after the sun rises, which helps to reset your circadian rhythms.

Stomach Settlers

As was discussed in Chapter 4, gastrointestinal distress is one of the leading problems experienced by ultrarunners during races. Even though training runs provide few opportunities to test the efficacy of any of the typical stomach settlers, such as ginger, antacids, and hard candies, you may be able to rule them out by testing them during training, even when your stomach is not upset. You are doing this to make sure your settler of choice doesn't cause any unforeseen issues.

GINGER

Ginger is available in candy form, crystallized, or in a ginger beer or brew. It has long been recommended by herbalists and doctors and used by cancer patients and pregnant women as an alternative to drugs to treat nausea. The theory is that the chemicals and soothing flavor help to calm an upset stomach. Although much of the research either shows a benefit or is inconclusive, small doses are reported to have the most benefit (Lien et al. 2003; Lohsirawat et al. 2010; Pongrojapaw, Somprasit, and Chanthasenanont 2007; Ryan 2012). You should note, however, that most of the generic ginger ales at aid stations contain no actual ginger!

ANTACIDS

These can be Tums or a generic alternative. When you eat, your body increases the amount of gastric acids in order to digest food. These acids can overflow up the esophagus and irritate and potentially damage the esophageal lining, as well as the lining of the stomach. To combat this increase in gastric acids, antacids provide a base in the form of calcium carbonate, which neutralizes the acid. Chew one or two at a time.

HARD CANDIES

Peppermints and butterscotch candies have long been used by runners to soothe sour stomachs. The theory is that the sucking motion of your mouth will relieve the distress. Although there is little scientific evidence to back up this claim, it's a home remedy that has been around for many years. In desperate times, it might be worth a shot. As an added bonus, you get the sugar from the candy. Butterscotch is my favorite.

This list of supplements and ergogenic aids is purposely very short. There should be no reason to incorporate anything other than real food, engineered sport-specific foods and sports drinks, and the items listed here. There are a lot of powders and packages in sports nutrition stores that promise to carry you farther and faster than ever before, but beware: supplements are not regulated by the Food and Drug Administration. They may contain ingredients that are not listed on the label, and the ingredients listed on the label may not be in the bottle or may be in the bottle in different amounts than listed. The combination of no regulation and inaccurate or deceptive manufacturing and labeling means that supplements from the corner store, grocery store, or specialized sports nutrition store may contain substances that are not only against the spirit of drug-free sport but are also prohibited by the US Anti-Doping Agency and World Anti-Doping Agency.

NSAIDs

Don't include NSAIDs (aspirin, ibuprofen, naproxen) in your race-day plan. Ever. They increase the risk of hyponatremia (Page et al. 2007; Wharam et al. 2006) and greatly increase the stress placed on the renal system, putting you at greater risk of renal injury and rhabdomyolysis (emptying of damaged or dead muscle tissue from muscles into the bloodstream). The desire to ache a little less isn't worth the risk of ending up in the hospital with renal failure. Pretty easy choice.

MODIFIED NUTRITION STRATEGY FOR RACE DAY

Table 16.1 showed a flawed, overcomplicated plan to demonstrate how not to develop a race-day nutrition plan. Now let's look at an example of what you should do. Table 16.2 shows a properly developed and streamlined race-day nutrition strategy. It is focused on target calorie and fluid ranges, not exact foodstuffs and specific scoops of powder. It gives the athlete the flexibility to choose to consume 100 calories of a prepackaged bar or the equivalent amount of energy from chews. The fluid is easy to manage, with target total consumption ranges from aid station to aid station. Finally, this is merely a replication of what the athlete would do in training. For a five-hour training run—similar to the five hours between aid stations 1 and 3—the athlete would take 80 to 120 oz of sports drink and water with 1,000 to 1,250 calories of gels, bars, rice balls, and energy chews.

RACE SECTION	FOOD	FLUID	SUPPLEMENTS
Start to aid station 1 (2 hours)	100 calories total (1 gel)	20-30 oz. total (water)	None
Aid station 1 to aid station 1 (2 hours)	400-500 calories total (gels and prepackaged bar)	30-50 oz. total (water and drink mix)	None
Aid station 2 to aid station 3 (3 hours)	600-750 calories total (rice balls, gels, energy chews)	50-70 oz. total (water and drink mix)	1 salt tab
Aid station 3 to finish (2 hours)	400-500 calories total (gels and prepackaged bar)	~50 oz. total (water and drink mix)	Ginger chews or antacid if necessary

TABLE 16.2 A simplified nutrition plan.

Recruiting and Instructing Your Support Crew

Ultrarunning has a special community. Nearly every weekend in cities, parks, and wildlands all across the country, people are organizing and running ultramarathons. Aid stations are stocked, and volunteers arise in the wee hours of the morning to take their places. With comprehensive aid station support at most ultras these days, it is certainly not necessary to have a crew for your race, but it sure is nice to see a familiar face.

Given their vital role in providing personalized aid and encouragement, choosing your crew to support you on your ultramarathon adventure is not a decision to take lightly. When you choose the correct crew mates, they can push you to greater heights. They will know exactly what to say and what to give you, and they may even be able to anticipate what you will need next. Alternatively, when you choose the wrong crew, they can drag you down. The wrong crew can drain your energy, make mistakes in execution, and, worst of all, cause a DNF when you are able to move forward.

THE FIRST STEP: DETERMINE WHETHER YOU WANT A CREW AT ALL

Both new and experienced runners sometimes choose to forego crews entirely. Most races make this possible with bountiful aid stations and more than enough helpful volunteers to keep you moving. If, for whatever reason, your preferred crew (family, friends, etc.) cannot make the event, don't fret. The race will most likely be able to take care of you.

If you decide you want a crew, your next move is to determine who exactly you want out there. Husbands, wives, kids, friends, and running partners can all make up a good crew. If they care for you and can get away for the weekend, chances are they would jump at the chance to help. However, just because your friends and loved ones can be part of your crew does not mean that they should be. The best crew members are the people who know you as a runner and as a person . . . and have only a small amount of sympathy. After all, it's going to be tough out there, so it's best to have someone in your corner who does not mind telling you to suck it up.

ALIGN YOUR CREW WITH YOUR GOALS FIRST, AND THEN INSTRUCT THEM ON THE TACTICAL DETAILS

First and foremost, your crew is there to assist you in achieving your goals; handing off gels or a jacket and making soup are merely the means to that end. I consistently see crews with folders, spreadsheets, and labeled baggies in the aid stations of ultramarathon events. Any kid off the street could look at a highlighted portion of a worksheet, pull the food and gear listed on it, and lay them out for the runner. You don't need a supportive, caring, and engaged crew member for that. And if that's all you empower your crew to do, you are not leveraging the people around you to enhance your performance. The error people make is having crew members who are well instructed on what to do but poorly instructed on the crew's overall goals, as well as the runner's outcome and process goals. Make no mistake, these overall goals are far more important than the number of potatoes a crew member hands you at mile 30. Beginner racers and crews tend to rely more on worksheets and minute-by-minute instructions because the structure provides confidence. Your goal, however, should be to progress to the point where, if you have properly instructed and empowered your crew, they can do the job armed with an index card, a duffel bag full of your gear and nutrition, and driving directions.

DO YOU WANT A PACER?

Many ultramarathons, particularly at the 100K and 100-mile distances, allow the use of pacers to accompany the runner. In theory, the pacers can provide motivational support and offer a level of safety for their runner and other runners in the field. As an added benefit, serving as a pacer is a great way to get an introduction into the sport. Most pacers are well intended, but some end up undoing their runner's race. Personality conflicts, goal misalignments, and a lack of preparation have waylaid many an ultrarunner's best-laid plans. Using a pacer or choosing to go solo is entirely a personal preference. The decision ultimately lies with the runner.

SUCCESS IS ALWAYS THE RUNNER'S RESPONSIBILITY

Crew or no crew, pacer or no pacer, the responsibility of finishing the race and achieving your goals is yours alone. When you have this default mindset, your crew and/or pacer(s) merely catalyze the process; they are not the linchpins in the operation. Success will ultimately be up to you, your training, and the determination you put forth on race day. When you are choosing a crew or a pacer, remember that it is always your responsibility as a runner to succeed. Your crew or a pacer can help you thrive on race day, but you should not rely on them for your success.

The ADAPT System for Problem-Solving

Everyone has a plan until they get punched in the mouth. —MIKE TYSON

If you do one single thing at a high enough intensity for long enough, every once in a while the shit hits the fan. As much as you have trained, as patiently as you have paced, as dialed as your race-day nutrition is, and as experienced as you are, you will eventually get punched in the mouth, and it is going to hurt. Your legs will feel like lead, your effort will feel unreasonable, you will start tripping over roots and rocks, and your stomach will be

in knots. If you are especially unlucky, these infirmities will all happen at once and last for many miles. Maybe it won't happen in your next race or the one after that, but if you remain in the ultramarathon game for a long enough time, lady luck's evil doppelgänger will eventually find you.

Ultramarathons are long enough that you have the opportunity to go through (many) highs and (hopefully fewer) lows. Some of the highs will be amazing; some of the lows will be excruciating. For many athletes, that's part of the attraction to the sport. However, having things go wrong does not necessarily mean your race is over. Fortunately, most ultramarathon cutoffs are generous enough that you can have a bad patch (or two) and still complete the event. And the more fit you are, the more bandwidth you'll have to get through a rough patch.

Believe me, I would much rather my athletes never have to experience bad patches in their races. Ideally, ultramarathoners would just run and eat and run, and run and eat and run some more, and never have any issues. If you play your cards right and prepare well, this is usually the case. Nonetheless, it's wise to prepare yourself for some tough times. As the British writer and former Prime Minister Benjamin Disraeli said, "I am prepared for the worst, but hope for the best."

I have developed five simple steps you can apply to get yourself out of the proverbial hole, regardless of the situation you're in. These steps form the easy-to-remember and appropriate acronym ADAPT, which also serves as a reminder of what you need to do at your lowest of lows:

- ACCEPT
- DIAGNOSE
- ANALYZE
- PLAN
- TAKE ACTION

ACCEPT

Accept things as they are. We all live in the present. In the exact moment that things deteriorate, you have to be present. Sure, you can hope that Scotty will beam you up to another place and another time where your stomach is not tied up in knots, but that isn't going to happen. So, accept the fact that things suck at the moment. Accept that your primary outcome goal might go out the window. Accept that you might be stuck in the aid station for the next hour or even longer, or that you might need to curl up and take a trail nap (your water bottle or hydration pack can make for an outstanding pillow, by the way). Accept how you are right now, however bad that might be, and get over it.

Emotion clouds judgment. It pulls an opaque veil over the situation, effectively rendering you incapable of rational thought and action. Acceptance of the situation allows you to move forward. When you reach the point of acceptance, you forget the past. The rock you just stubbed your toe on for the fifth time? Gone. The trail marker you missed, costing you precious time and energy on the wrong path? In the past. You can't change the fact that you've tripped and fallen three times over the last hundred yards, but you can change your outlook. Acceptance of

the situation moves you from the past into the present. It lifts the veil of emotion and enables you to think and act rationally. Accept first, and then you are ready to move forward.

DIAGNOSE

Make a quick and dirty assessment of what is going on. Don't try to solve your problems yet, just try to figure out what is going on. This step is easy. If you just rolled your ankle, then, duh, you have an injured ankle. If your stomach has turned, then your nutrition plan has gone awry. If you are frustrated because you have fallen, then you are simply frustrated. Don't worry about the specifics of the issue just yet or how to resolve it; just diagnose the problem. Keep this step simple and to the point. "I have an upset stomach," "I am lightheaded," or "I am frustrated" (or a combination of these) will work perfectly fine. It is also fine to identify more than one problem, but resist the urge to roll right to analysis or planning before clearly identifying the problem. When people fail at this step, they end up creating solutions to the wrong problems.

ANALYZE

Now it is time to apply some brain power to the problem and enlist whatever synapses you have that are still working. You have moved on and accepted that things suck. You have diagnosed what the issue is. It's now time to analyze the situation you are in. Where is the next aid station? How much time do you have until the next cutoff? What tools, food, supplies, and gear are available? Create a mental inventory, because these are the means you will use to get yourself out of the hole you are currently in. The outcome of the next step depends on the analysis you do!

PLAN

You have accepted the situation, diagnosed what is wrong, and analyzed your surroundings and tools. Now it's time to actually devise a plan. This is by far the most complicated step. The plan should not require Mensa-level analysis, but it will require some brainpower. Your plan incorporates your earlier analysis of the situation and the means at your disposal. It takes the wheres and whats and weaves them into concrete steps that can lift you out of the hole you have dug. Depending on the situation, a simple plan might be to get to the next aid station and figure it out from there. If this is the case, you can share the results of your diagnosis and analysis with your crew, and they can help you formulate a new plan. One step at a time.

TAKE ACTION

When it's all said and done, you have to take action. Problems do not fix themselves. You as the athlete have to do something deliberate to fix them. If you believe in magic, like the Disneyland type of magic, then ultrarunning is not for you. Put your plan into action. Take action—by force if necessary.

Situation: "I have just rolled my ankle on a rock."

Accept: “My ankle is going to hurt for a bit. I am going to be slower. This is fine. I’m over it.”

Diagnose: “I have an injured ankle.”

Analyze: “I was sixty minutes away from the next aid station. Now I am about ninety minutes away if I walk. I do not have enough food or water on me for that length of time. I have crew at the next aid station.”

Plan: “I am going to walk into the next aid station. If I see another runner, I will ask them for some food and fluid. When I get to the next aid station, I am going to see if there is medical help there or some other way to tape/brace my ankle. My crew is there, so they can help me with this.”

Take action: “I am going to walk down the trail now.” As another runner approaches: “I am going to ask this runner for some fluid.”

WHAT IS MISSING FROM THE ADAPT SYSTEM?

Admittedly, the acronym ADAPT is a bit contrived. It represents a series of steps that are analogous to problem-solving techniques that have been used by Boy Scouts, mountaineers, adventure racers, and the military, among others. The point of using the acronym is to identify a series of concrete steps you can focus on when you are at the lowest of low points. It is also important to understand what I intentionally left out of the ADAPT system.

Predict

Many runners are by nature analytical people. They are experts at doing easy math and determining their pace and how long it is going to take to get to the next aid station. While under normal circumstances it is a good thing to know when and how long it will take to locomote from place to place, when one is trying to climb out of a negative situation, it’s best to keep that thinking to a minimum. You should absolutely figure out how long it is going to take you to get to the next aid station, but the math should end there; do not forecast the following aid station, or the one after that, and certainly not the finish line. One never knows how the day is going to turn out.

Radically Change Your Race Strategy

The acronym is ADAPT, not PANIC. Making small, incremental changes is always better than drastically revising your well-thought-out race strategy all at once. Many times, small changes (or adaptations) to your original plan are all that are necessary.

Guess

Simply taking a stab in the dark to fix a problem should be the last resort. If you do a thorough job of analyzing and planning, the steps to get out of the hole should be quite clear-cut. Fortunately for most of us, fixing problems in ultrarunning is not rocket science. There is typically a wealth of experience out on the course during race day. Your crew, aid station workers, and fellow runners can help if you get stumped. Use them if necessary!

KEY POINTS FOR PERSONALIZING YOUR RACE STRATEGY

Personalizing your race-day strategy starts with reviewing your training.

Race-day strategies should consist of both process and outcome goals.

Race-day strategies should balance your current fitness levels with your affinity for risk.

Your race-day strategy should be based on effort (as opposed to heart rate or pace) and be constructed from your training experience.

Beware of forecasting how you will feel during a race. Staying in the moment works!

Longer races such as 200-milers will require special strategies around sleep:

- **Pre-event “sleep banking” can be effective.**
- **Sleep deprivation training is likely only beneficial for the experiential aspect.**
- **Sleeping during longer races should prioritize preserving performance over limiting downtime.**

Race-day nutrition strategies should be simple and based on what you have done in training.

You should try supplements and other ergogenic aids like ginger and caffeine in training before you try them on race day.

When shit hits the fan, ADAPT: Accept, Diagnose, Analyze, Plan, Take action.

